

HDOC - DAY 10

C Programming Activity

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SUBJECT: C programming

DATE: 3-09-26

Question 1

Write a C program to input time in seconds and convert it into days, hours, minutes, and seconds format.

1. Aim

To write and execute a C program that accepts time in seconds and converts the entered value into days, hours, minutes, or seconds according to the user's selected option.

2. Algorithm

1. Start the program.
2. Declare float variables b, c, d and integer variables a, i.
3. Read the time value i in seconds.
4. Display the four conversion choices and read choice a.
5. Calculate $b = i / 60$, $c = i / 3600$, and $d = i / (24 \times 3600)$.
6. If $a = 1$, display total days; if $a = 2$, display total hours; if $a = 3$, display total minutes; if $a = 4$, display total seconds.
7. Stop the program.

3. Pseudocode

```
START
INPUT i
DISPLAY conversion menu
INPUT a
 $b \leftarrow i / 60$ 
 $c \leftarrow i / 3600$ 
 $d \leftarrow i / (24 \times 3600)$ 
IF  $a = 1$  THEN DISPLAY d
ELSE IF  $a = 2$  THEN DISPLAY c
ELSE IF  $a = 3$  THEN DISPLAY b
ELSE IF  $a = 4$  THEN DISPLAY i
END IF
STOP
```

4. Flowchart



5. Test Case Table

Test Case	Input Seconds	Choice	Expected Output	Result
1	86400	1	1.000000 days	Pass
2	7200	2	2.000000 hours	Pass
3	3600	3	60.000000 minutes	Pass
4	125	4	125.000000 seconds	Pass
5	172800	1	2.000000 days	Pass

6. C Program

```
#include<stdio.h>
int main()
{
    float b,c,d,e;
    int a,i;
    printf("Enter Seconds: ");
    scanf("%d",&i);

    printf("\n1. Convert it into Days: \n");
    printf("2. Convert it into Hours: \n");
    printf("3. Convert it into Minutes: \n");
    printf("4. Convert it into Seconds: \n");
    scanf("%d",&a);

    b=(float)i/60;
    c=(float)i/3600;
    d=(float)i/(24*3600);
    e=(float)i;

    if(a==1)
        printf("Total Days: %f",d);

    else if(a==2)
        printf("Total Hours: %f",c);

    else if(a==3)
        printf("Total Minutes: %f",b);

    else if(a==4)
        printf("Total Seconds: %f",e);

    return 0;
}
```

7. Compilation, Execution and Testing

```
$ gcc time.c -o time
$ ./time
Enter Seconds: 86400
```

```
1. Convert it into Days:
2. Convert it into Hours:
3. Convert it into Minutes:
```

4. Convert it into Seconds:

1

Total Days: 1.000000

```
(kali㉿kali)-[~/Desktop/cprog]
$ gcc leapyear.c -lm
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
To check if a year is a Leap!
Enter a year: 2024
It is a Leap year
```

```
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
To check if a year is a Leap!
Enter a year: 2025
Not a Leap year
```

```
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
To check if a year is a Leap!
Enter a year: 2000
It is a Leap year
```

```
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
To check if a year is a Leap!
Enter a year: 1800
Not a Leap year
```

```
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
To check if a year is a Leap!
Enter a year: 1900
Not a Leap year
```

Question 2

Write a C program to input a year and check whether it is a leap year or not.

1. Aim

To write and execute a C program that accepts a year as input and checks whether it is a leap year or not.

2. Algorithm

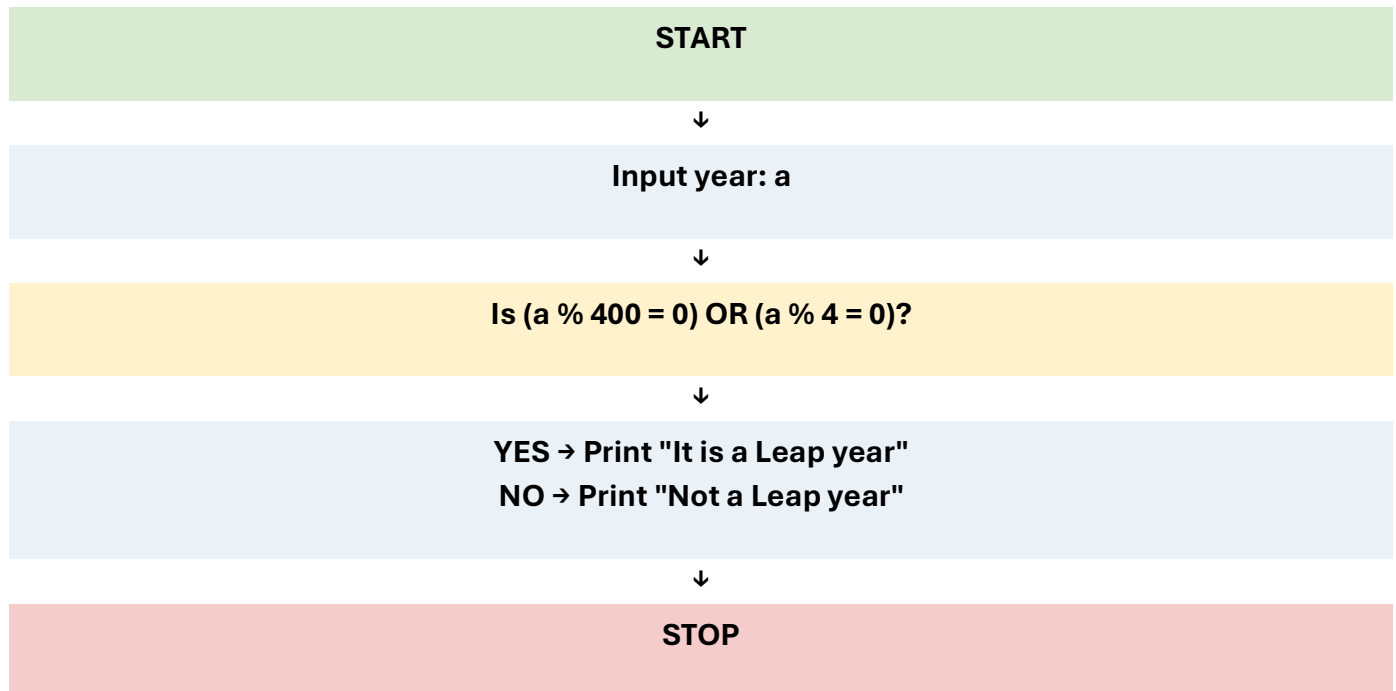
1. Start the program.

2. Declare integer variables a, b, i, j.
3. Input the year in variable a.
4. Check whether the year is divisible by 400 or divisible by 4.
5. If the condition is true, display “It is a Leap year”.
6. Otherwise, display “Not a Leap year”.
7. Stop the program.

3. Pseudocode

```
START
INPUT a
IF (a MOD 400 = 0) OR (a MOD 4 = 0) THEN
    DISPLAY "It is a Leap year"
ELSE
    DISPLAY "Not a Leap year"
END IF
STOP
```

4. Flowchart



5. Test Case Table

Test Case	Input Year	Correct Expected Result	Output from Screenshot Code	Result
1	2024	Leap year	It is a Leap year	Pass

2	2025	Not a leap year	Not a Leap year	Pass
3	2000	Leap year	It is a Leap year	Pass
4	1800	Not a leap year	It is a Leap year	Fail*
5	1900	Not a leap year	It is a Leap year	Fail*

***Important:** The screenshot code checks divisibility by 400 OR 4 only. Therefore century years such as 1800 and 1900 are incorrectly reported as leap years. The code below is kept the same as the screenshot, as requested.

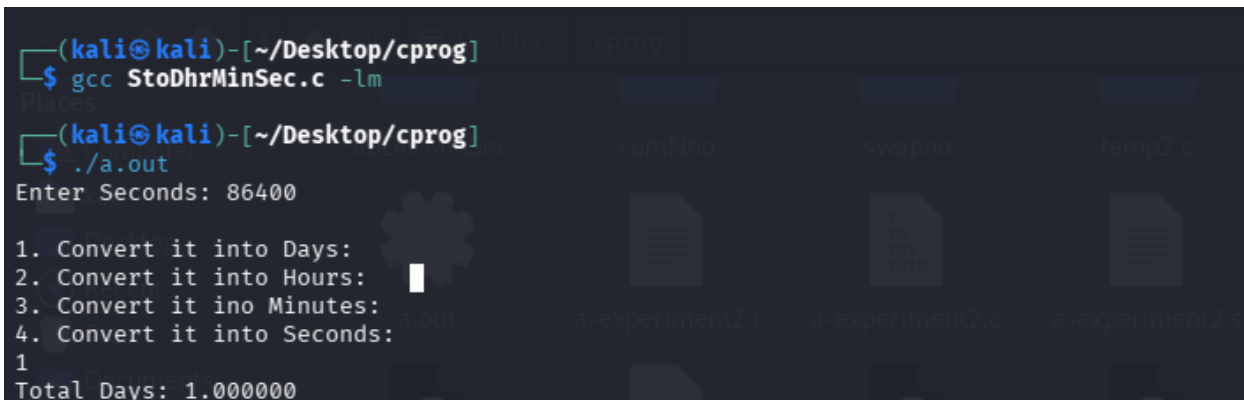
6. C Program

```
#include<stdio.h>
int main()
{
    int a,b,i,j;
    printf("To check if a year is a Leap!\n");
    printf("Enter a year: ");
    scanf("%d",&a);

    if((a%400 == 0) || (a%4 == 0))
        printf("It is a Leap year");
    else
        printf("Not a Leap year");
}
```

7. Compilation, Execution and Testing

```
$ gcc leapyear.c -o leapyear
$ ./leapyear
To check if a year is a Leap!
Enter a year: 2024
It is a Leap year
```



```
(kali㉿kali)-[~/Desktop/cprog]
$ gcc StoDhrMinSec.c -lm
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter Seconds: 86400
1. Convert it into Days: 1
2. Convert it into Hours: 24
3. Convert it into Minutes: 1440
4. Convert it into Seconds: 86400
Total Days: 1.000000
```

```
(kali㉿kali)-[~/Desktop/cprog]
```

```
$ ./a.out
```

```
Enter Seconds: 7200
```

1. Convert it into Days:
2. Convert it into Hours:
3. Convert it into Minutes:
4. Convert it into Seconds:

```
2
```

```
Total Hours: 2.000000
```

```
(kali㉿kali)-[~/Desktop/cprog]
```

```
$ ./a.out
```

```
Enter Seconds: 3600
```

1. Convert it into Days:
2. Convert it into Hours:
3. Convert it into Minutes:
4. Convert it into Seconds:

```
3
```

```
Total Minutes: 60.000000
```

```
(kali㉿kali)-[~/Desktop/cprog]
```

```
$ ./a.out
```

```
Enter Seconds: 125
```

1. Convert it into Days:
2. Convert it into Hours:
3. Convert it into Minutes:
4. Convert it into Seconds:

```
4
```

```
Total Seconds: 125.000000
```

```
(kali㉿kali)-[~/Desktop/cprog]
```

```
$ ./a.out
```

```
Enter Seconds: 172800
```

1. Convert it into Days:
2. Convert it into Hours:
3. Convert it into Minutes:
4. Convert it into Seconds:

```
1
```

```
Total Days: 2.000000
```